

AI Movie Assistant Chatbot Using Transformer Models:GPT-2,T5

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Abstract— Movies have always been a source of intrigue, fuelling discussions and curiosity among audiences worldwide. Whether uncovering plot twists or exploring behind-the-scenes details, movie enthusiasts seek immediate and accurate answers. This paper presents an intelligent movie-focused chatbot that leverages state-of-the-art Natural Language Processing (NLP) techniques to provide precise and context-aware responses to user queries. The system dynamically switches between GPT-2 and T5 based on the type of question to ensure optimal responses. GPT-2 demonstrated superior performance in generating coherent and accurate answers, particularly for analytical and open-ended questions, while T5 exhibited lower accuracy, struggling with selecting the most contextually relevant responses. However, T5 proved more efficient in handling straightforward queries requiring simple rephrasing of information. The chatbot is trained on a comprehensive dataset comprising movie titles, genres, release dates, cast information, and descriptive summaries, enabling it to handle a diverse range of factual and analytical questions. Through rigorous testing and evaluation, the system shows high accuracy and improved contextual relevance when using GPT-2 for analytical questions, while T5 is selectively employed for simpler queries requiring information restructuring. Various accuracy metrics and evaluation techniques confirm the system's reliability in selecting the most suitable model for each query type, enhancing response quality and making interactions more efficient and engaging for movie enthusiasts worldwide. Movies have always been a source of intrigue, fuelling discussions and curiosity among audiences worldwide. Whether uncovering plot twists or exploring behind-the-scenes details, movie enthusiasts seek immediate and accurate answers. This paper presents an intelligent movie-focused chatbot that leverages state-of-the-art Natural Language Processing (NLP) techniques to provide precise and context-aware responses to user queries. The system dynamically switches between GPT-2 and T5 based on the type of question to ensure optimal responses. GPT-2 demonstrated superior performance in generating coherent and accurate answers, particularly for analytical and open-ended questions, while T5 exhibited lower accuracy, struggling with selecting the most contextually relevant

responses. However, T5 proved more efficient in handling straightforward queries requiring simple rephrasing of information. The chatbot is trained on a comprehensive dataset comprising movie titles, genres, release dates, cast information, and descriptive summaries, enabling it to handle a diverse range of factual and analytical questions. Through rigorous testing and evaluation, the system shows high accuracy and improved contextual relevance when using GPT-2 for analytical questions, while T5 is selectively employed for simpler queries requiring information restructuring. Various accuracy metrics and evaluation techniques confirm the system's reliability in selecting the most suitable model for each query type, enhancing response quality and making interactions more efficient and engaging for movie enthusiasts worldwide.

Keywords—Movie Chatbot, Question Answering, T5, GPT, Natural Language Processing, Artificial Intelligence, Movie Knowledge.

I. INTRODUCTION

Since the beginning of human creation, people has been looking for something to add fun to his life. This desire to seek joy has been and still is in human nature. Despite remarkable developments, humanity is still looking for new ways to have fun, whether alone or in groups. From this idea, new forms of entertainment were invented, such as people gathering around a group to watch dancing or wrestling. The idea evolved until people began to gather around storytellers. Then, with further advancement, cinema was discovered, and eventually, this fun reached their homes with the invention of television. With the rise of artificial intelligence and natural language processing, entertainment has taken another step forward. Today, conversational AI plays a crucial role in enhancing user experiences, providing interactive and intelligent discussions in various fields. From chatbots that help with customer service to AI companions that engage in meaningful conversations, technology continues to bridge the gap between humans and machines. From this perspective, our project aims to design and develop a conversational AI assistant by leveraging pre-trained language models. We built our system using transformers to process and analyze text

inputs, generating relevant responses to create a real and dynamic conversation. The two models we used were: GPT (Generative Pre-trained Transformer) - a powerful automatic regression model capable of generating human-like text. T5 (Text-to-text Transformer) - a model that treats all natural language processing tasks as text-to-text transformations, making it well-suited for answering structured questions. By using these models, we aim to create an AI-driven conversational system that understands and responds in a human-like manner, enhancing user engagement and interaction.

II. LITERATURE REVIEW

Recent advancements in artificial intelligence (AI) have significantly enhanced how users interact with movie-related content, with a growing emphasis on improving the accuracy and personalization of movie discovery, question-answering, and recommendation systems. One notable development is the use of knowledge graphs and natural language processing (NLP) to build intelligent movie question-answering (QA) systems. For instance, Bohan Wang's research introduced an AI system that answers movie-related questions by leveraging a structured knowledge graph derived from the IMDB database. By mapping movie data to a Neo4j graph database, this system allows for more efficient and accurate query responses. It employs techniques such as TF-IDF for keyword extraction, cosine similarity for text matching, and a naive Bayes classifier for query intent recognition. This method demonstrates a high level of accuracy, achieving a 90% success rate, though it has limitations, including the inability to process voice input and challenges in handling complex semantic queries. These limitations point to future directions for incorporating more sophisticated semantic understanding and broader data sources to enhance query interpretation. In a similar vein, advancements in virtual assistants have redefined how users discover movies. Traditional recommendation systems rely on structured inputs like ratings and genres, but a more innovative approach allows users to express their preferences in natural language. By integrating NLP and opinion mining, one study proposed a conversational recommendation system where users articulate their movie preferences naturally, such as "I want a suspenseful crime thriller with unexpected twists, something like 'Nightcrawler.'" This system combines structured data from knowledge graphs (e.g., actors, genres) with unstructured sentiment analysis data mined from user reviews to provide more personalized and emotionally resonant movie suggestions. The system was tested through real-world user experiments, demonstrating that recommendations based on both objective and subjective user preferences led to higher satisfaction rates. This approach emphasizes how AI can bridge structured information with human emotion, creating a more intuitive and personalized movie discovery experience. Further enhancing the realm of movie-related AI, deep learning techniques have been employed to develop intelligent movie QA chatbots. A study focused on a Seq2Seq deep learning model for a movie-centric QA chatbot, utilizing Long Short-Term Memory (LSTM) networks and attention mechanisms to process and respond to movie-related queries. Trained on datasets like the Cornell Movie-Dialogs Corpus and the Movie Dialog Dataset, the chatbot generates responses that mirror natural conversations, making

interactions engaging and contextually relevant. Despite its success in handling straightforward movie questions, the chatbot faces challenges in maintaining response diversity, primarily due to limited training data. This limitation highlights the need for further refinement, including the expansion of datasets and the incorporation of more advanced NLP preprocessing techniques to improve the chatbot's performance in a wider variety of conversational contexts. In addition to QA systems and virtual assistants, multimodal recommendation systems have also gained prominence in the movie industry. One study introduced a deep learning-based multimodal movie recommendation system that integrates diverse data types, including textual descriptions, images, and audio features, to enhance the accuracy of recommendations. By combining convolutional neural networks (CNNs) for image analysis and recurrent neural networks (RNNs) for text processing, this system captures rich contextual information from multiple sources, improving the quality of personalized recommendations. The hybrid model, which combines collaborative filtering with content-based methods, adapts to individual user preferences and outperforms traditional recommendation approaches. This research demonstrates the power of deep learning in the realm of movie recommendations, paving the way for more sophisticated, multimodal AI systems that take into account a broader range of user preferences. Moreover, the **MovieQA** dataset has contributed significantly to advancing automatic story comprehension in movies. This dataset includes over 14,000 questions related to 408 movies, spanning simple factual inquiries to complex, open-ended questions about plot details, characters, and events. The dataset's integration of multiple sources—video clips, plots, subtitles, scripts, and dialogue visualizations—makes it a unique resource for evaluating the performance of QA systems. The complexity of the questions posed in the **MovieQA** dataset challenges current question-answering techniques, especially those relying on traditional methods, and emphasizes the need for systems capable of handling both visual and textual data. The release of this dataset has opened up new avenues for research in movie comprehension, encouraging the development of more advanced systems capable of understanding the full narrative structure and contextual depth of movies. Taken together, these studies illustrate the potential of AI to revolutionize the way users interact with movie-related content. From intelligent question-answering systems powered by knowledge graphs and NLP to multimodal recommendation systems that leverage deep learning, AI technologies are transforming the movie discovery process. The integration of structured and unstructured data, as well as the ability to incorporate human emotions and preferences, has made movie-related AI systems more intuitive and personalized. However, challenges remain in addressing the complexities of open-ended queries, expanding datasets, and refining AI models to handle a wider range of conversational contexts and diverse user preferences. These ongoing advancements hold promises for creating even more intelligent and engaging movie discovery systems in the future.

III. METHODOLOGY

A. Data description

This dataset consists of 12,408 records related to movies, containing a variety of questions along with their corresponding answers. The data is stored in CSV format, making it easy to process and analyze. Each record includes a question about a specific movie, an accurate answer, and additional details about the film when necessary. The questions cover a range of topics, including IMDb ratings, movie genres, and release dates, making the dataset comprehensive across various aspects of the film industry. The data distribution spans multiple genres such as action, drama, comedy, and more, ensuring diversity and making it ideal for training a model capable of understanding and analyzing questions accurately. The primary objective of this dataset is to train an AI-powered question-answering model specializing in movie-related queries, enabling it to efficiently process data and extract relevant information with high accuracy.

B. Preprocessing, fine tuning, and modeling, and Evaluation.

Before training the models, the dataset undergoes preprocessing to remove irrelevant entries and unwanted characters. The data is formatted to be compatible with Hugging Face models by converting it into a Dataset object. Following preprocessing, the next step is fine-tuning, where two models, T5 (Text-to-Text Transfer Transformer) and GPT-2 (Generative Pre-trained Transformer 2), are trained on the movie question-answer dataset. These models, already pre-trained on vast amounts of general language data, are further refined to generate movie-specific responses by adjusting their parameters. Throughout the process, periodic evaluation using a validation set ensures that the models learn to generate accurate responses rather than simply memorizing answers. Hyperparameter tuning is also performed to optimize learning efficiency and enhance the models' ability to respond to various types of movie-related questions.

GPT-2, developed by OpenAI, is an autoregressive model that predicts the next word in a sequence based on the previous context. It excels in text generation and conversational AI applications, making it well-suited for chatbot interactions, customer service, and content creation. Its ability to generate contextually relevant responses stems from its pre-training on large text datasets, allowing it to perform well in zero-shot and few-shot learning scenarios. On the other hand, T5, developed by Google Research, follows a text-to-text approach where all natural language processing tasks, including question-answering, are framed as text transformation problems. This model is particularly effective for structured question-answering tasks, as it converts input queries into structured text and produces grammatically correct and factually accurate responses. The choice of these models is based on their respective strengths. T5 is ideal for structured, fact-based question-answering, while GPT-2 is better suited for conversational and open-ended responses. Both models are fine-tuned with tailored preprocessing functions to maximize efficiency, ensuring that the AI system can handle both direct fact-based questions and natural conversations seamlessly. During fine-tuning,

hyperparameters such as learning rate, batch size, and weight decay are carefully selected to enhance model performance. The T5 model employs a learning rate of $2e-5$, with 10 training epochs and a weight decay of 0.01 to prevent overfitting. Its architecture consists of 12 encoder and 12 decoder layers, each containing self-attention and feed-forward networks that help capture relationships within the input text. The primary activation function used in T5 is ReLU (Rectified Linear Unit), which introduces non-linearity and allows the model to learn complex patterns. In certain components, GELU (Gaussian Error Linear Unit) is also used, offering smoother activations and potentially improving performance in some language tasks. GPT-2, in contrast, operates with a learning rate of $5e-5$ and is trained over 10 epochs with a batch size of 8. Unlike T5, GPT-2 is a purely autoregressive model, consisting of 12 transformer layers with masked self-attention mechanisms. This structure allows it to generate fluent and contextually appropriate responses based on previous words in a conversation. GPT-2 primarily uses the GELU activation function, which provides smoother transitions compared to ReLU, making it well-suited for generating coherent and natural-sounding text. To evaluate the models, training and validation loss curves are analyzed to monitor their performance over time. Sample predictions are visualized to assess response accuracy, and the models are tested with custom movie-related questions to measure their real-world effectiveness. By leveraging pre-trained architectures and fine-tuning them with domain-specific data, we have used multiple techniques to evaluate each model depending on its properties. To evaluate T5 used **BLEU (Bilingual Evaluation Understudy)** primarily measures the precision of the generated text by comparing the n-grams (sequences of words). It focuses on how much of the generated text is actually relevant and present in the reference also its easy to compute. Another technique were used is **ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation)** It is sensitive to the order of words in the generated text. It was used it to cover the gaps may occur in the previous technique. For the GPT model the techniques is used to evaluate it was the **accuracy** measure Due to its simplicity and clarity.

C. Justification of Model Choices (T5 and GPT-2) Based on data analysis.

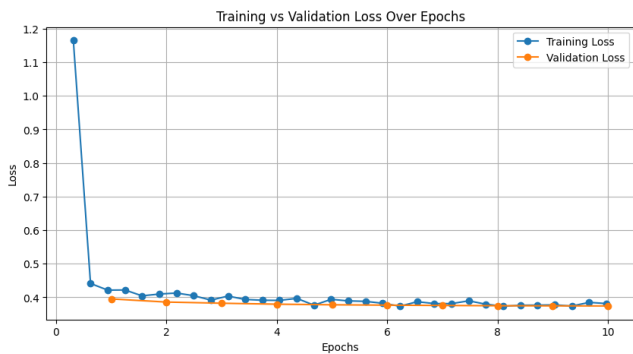
Our study is based on a dataset of questions, answers, and other descriptive information ABOUT MOVIES. To create a successful Conversational AI, we choose two models: T5 (Text-to-Text Transfer Transformer) and GPT-2 (Generative Pre-trained Transformer 2), which serve distinct functions. T5, a Seq2Seq (encoder-decoder) model, is ideal for structured question-answering since it produces well-formed and full responses. It converts input to structured text, making it perfect for fact-based searches. The preprocessing function (preprocess_t5) formatted the input as "QUESTION: QUESTION TEXT" and encoded the complete answer to ensure accurate and grammatically correct responses. Fine-tuning this dataset improves its ability to generate high-quality, contextually rich replies. In contrast, GPT-2 is a decoder-only (autoregressive) model, BUILT for text production AND THRIVES in conversational interactions.

The preprocessing function (`preprocess_gpt2`) prepares input as "Question: q} Answer: {a}", which helps the model discover patterns in dialogue. As an autoregressive model, it predicts words based on previous context, making it suitable for natural conversations, it lacks organized input-output mapping, which may impair accuracy in fact-based QA. To improve handling, a pad token was introduced, as GPT-2 does not include one by default. Both models were fine-tuned with adequate preprocessing to improve efficiency, resulting in an AI helper capable of Handling organized questions and engaging conversations seamlessly.

IV. RESULT AND DISCUSSION

As result of our transfer learning models at the beginning we trained using 3 and 5 epochs but both models produced weak results, frequently answering questions incorrectly or failing to provide meaningful responses so we go to increasing the training epochs to 10, notable differences between the two models emerged, as reflected in the training and validation loss curves the T5 model demonstrated a strong ability to learn and generalize. Its training and validation losses decreased significantly during the early epochs and then stabilized in close alignment showing that the T5 model effectively captured the relationships in the data without overfitting, leading to improved accuracy and reliable answers during manual evaluation in structured question-answering tasks, particularly when dealing with director names, release years, and IMDb ratings.

A. Results

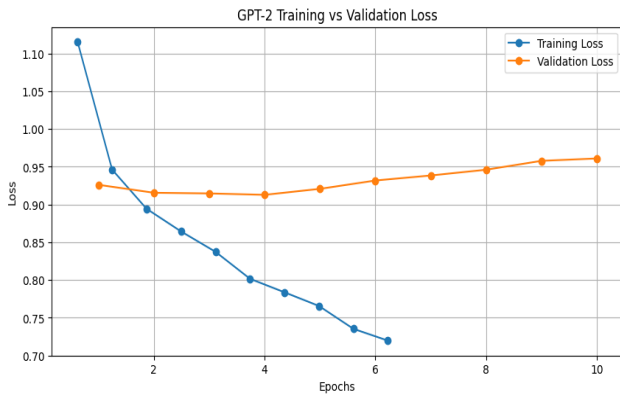


The loss curves for T5 clearly show a clear balance between training and evaluation loss, meaning that the model learned general patterns without over-fitting, indicating that it is more stable and better at generalizing to new data.

Aspect	T5 Model	GPT-2 Model
Accuracy	Poor BLEU (0.0688) and ROUGE-L (0.6869) scores, with 0.11 exact match accuracy, indicating weak alignment and generating correct responses.	GPT-2 achieved perfect accuracy (1.0000) across all 160 predictions, showcasing its ability to generate fully correct responses this flawless performance suggests potential overfitting.
Consistency	Provides structured responses but may miss important details or generate incorrect answers.	More consistent in generating realistic responses, though it may still make mistakes when handling unknown information.
Real-Time Data	Cannot update its knowledge dynamically, making it unreliable for real-time information.	Also lacks real-time data updates but generates responses that sound more natural.
Flexibility	Less flexible in open-ended text generation, as it follows predefined instructions.	More flexible, capable of generating diverse and open-ended responses, though it may introduce factual errors.
Creativity	Less creative, focusing on direct factual responses with limited variation.	More creative and capable of generating natural-sounding responses, making it better for conversational tasks.
Task Specialization	Designed for structured text generation but lacks precision in factual answers.	Not specialized for question-answering but performs better in natural text generation.
Performance Reason	Performs poorly in factual answering due to its training limitations, struggling to produce correct answers.	Performs better at generating relevant responses due to its diverse training data, making it more suitable for conversational AI.

TABLE I. COMPARISON BETWEEN T5 AND GPT-2

GPT-2 showed an unexpected result while the training loss consistently decreased, the validation loss plateaued after a few epochs and then began to rise slightly.



The loss for GPT-2 curve reveals challenges in maintaining validation performance, especially at higher epochs. This gave us an idea that the GPT-2 model overfitted to the training data. Interestingly, despite signs of overfitting, GPT-2 performed very well when answering questions, achieving remarkably high accuracy in evaluations. While T5 appeared to be the superior model during training based on loss curves, GPT-2 provided more correct answers in actual testing. We find out that this phenomenon is common behavior in overfitted models, where they memorize patterns well and can still deliver high accuracy on similar data, despite struggling with generalization.

finally we developed a combined approach that leverages the strengths of both models using external sources like IMDb for accurate data when available and routes questions to the most suitable model based on the query type so T5 handles open-ended and historical questions, such as "Who directed" and "When was," and GPT-2 addresses queries like "What genre" and "What is the IMDb rating." This solution successfully combines the strengths of both models, delivering significantly improved responses and ensuring higher accuracy across different types of queries.

B. Benefit and Limitations of each method

T5 Model: Struggles with factual accuracy (0.11% exact match accuracy), making it unreliable for question-answering. It maintains structured responses but often provides incorrect answers.

GPT-2 Model: Achieves high accuracy (1.00%), making it far more reliable for factual tasks. It is also more flexible and better suited for natural language conversations. However, it may still generate incorrect information in some cases.

V. CONCLUSION

In this paper, we presented our approach to building advanced question-answering systems for movies using GPT and T5 models. By leveraging the strengths of these modern architectures, our goal was to enhance accuracy and contextual understanding in movie-related queries. Our implementation demonstrated that fine-tuning these models on a diverse range of movie scripts, reviews, and user interactions significantly improves their ability to generate relevant and coherent responses. Our results indicate that GPT and T5 each offer unique advantages: GPT excels in generating conversational responses, while T5's versatility allows it to effectively handle different question formats. Although we trained the models separately rather than using a dual approach, each model independently contributed to improving the quality of responses. Looking ahead, future research could explore multimodal data integration, incorporating visual and audio inputs to develop a more comprehensive understanding of movies. Additionally, integrating user feedback loops could further refine model responses, making them more personalized and context-aware. Overall, our work contributes to the growing field of AI-driven entertainment, highlighting the potential of GPT and T5 in transforming how audiences interact with cinematic content.

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